Fundamentals of AI and KR - Module 3

1. Introduction to uncertainty and probabilistic reasoning

Roberta Calegari

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Introduction

Problem-solving logical agents

- Consider 3 switches in a switchboard: A, B, and C. When A is on, B is also on. When C is on, B is off. If both A and C are on, the switchboard melts down. *Can that ever happen?*

Introduction

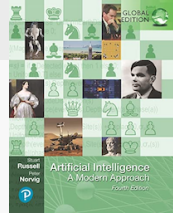
Problem-solving logical agents

- Consider 3 switches in a switchboard: A, B, and C. When A is on, B is also on. When C is on, B is off. If both A and C are on, the switchboard melts down. *Can that ever happen?*
- There are 12 coins distributed in piles of 4, 1, and 7 coins. We can move coins from a pile A to a pile B, but only by doubling the coins in B. *Can we distribute coins evenly among the three piles?*

Introduction

Problem-solving logical agents

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- We're guests on a TV game show. We stand in front of three closed doors. A prize hides behind one of them. We choose the door on the left. At this point, the host, who knows where prize is, opens the middle door, to reveal it is empty. We are offered to modify our choice. *Should we?*



Handling uncertainty

Agents operate under uncertainty when they face...

- **Partial observability:** relevant information missing, delayed, noisy...
- **Uncertain outcomes:** the same action may lead to different results
- **Dynamic environments:** world may change while reasoning

Examples: autonomous driving, medical decision support, fraud detection,...

Logical agents can represent alternative states and contingency plans.
However, in realistic settings:

- the number of possible states grows rapidly;
- exhaustive contingency plans become impractical;
- no action can guarantee success.

Probability

Probability provides a compact way to represent uncertainty, update beliefs from evidence, and support decisions under risk.

Topics

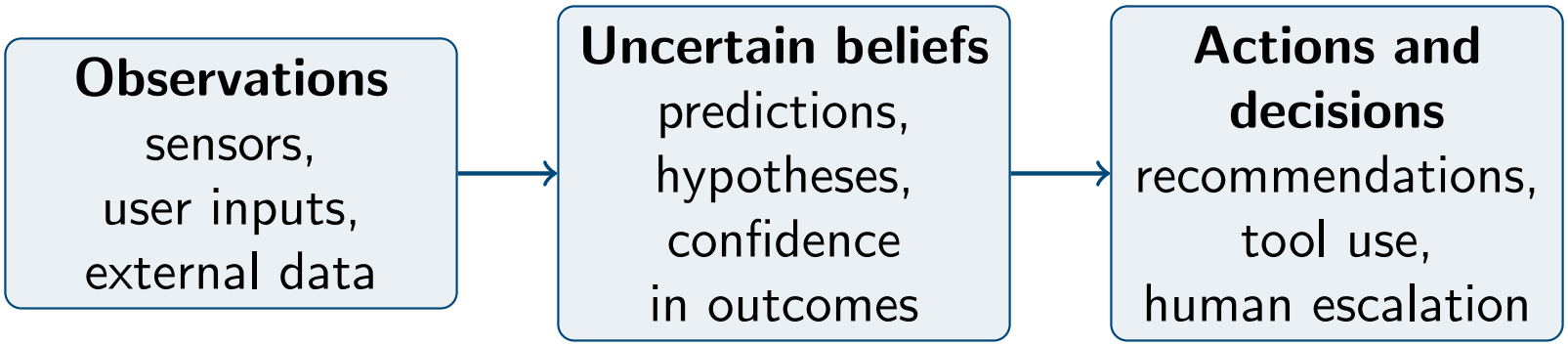
- Basic probability notation
- Conditional probability and Bayes' rule
- Inference using full joint distributions
- Independence and conditional independence
- Bayesian network representation
- Constructing Bayesian networks
- Exact and approximate inference
- Simple case studies

Final goal

By the end of the module, you will be able to model uncertainty, update beliefs from evidence, and use probabilistic inference to support informed decisions.

Why probability still matters in modern AI

Modern AI systems produce useful predictions/ recommendations, but they **must** often operate with **incomplete**, **noisy**, or **ambiguous** information.



Probability is not an alternative to modern AI. It provides a principled language to represent and update uncertainty about an AI system's inputs, predictions, and possible consequences.

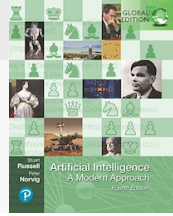
Examples

Autonomous driving, medical decision support, fraud detection, and AI assistants that use external tools all need to reason under uncertainty.

Acting under uncertainty

Key idea

An intelligent agent must often choose an action without knowing its outcome with certainty. Probability represents beliefs about uncertain outcomes; utility represents preferences over their consequences.



Uncertainty

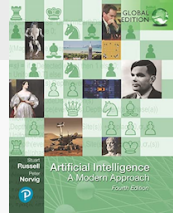
We need to reach the airport on time. Let action A_t = leave for airport t minutes before flight. *Will A_t get me there on time?*
Problems:

1. partial observability (road state, other drivers' plans, etc.)
2. noisy sensors (traffic reports)
3. uncertainty in action outcomes (flat tire, etc.)
4. immense complexity of modelling and predicting traffic

Hence a purely logical approach either

1. risks falsehood: *"A_{25} will get me there on time,"* or
2. leads to conclusions that are too weak for decision making: *"A_{25} will get me there on time if there's no accident on the bridge, and it doesn't rain and my tires remain intact etc etc."*

(A_{1440} might reasonably be said to get me there on time but I'd have to stay overnight in the airport . . .)



Methods for handling uncertainty I

Default or nonmonotonic logic

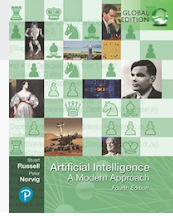
- Assume my car does not have a flat tire
- Assume A_{25} works unless contradicted by evidence

What assumptions are reasonable?

Rule-based systems with heuristic confidence weights (empirically motivated)

- $A_{25} \mapsto_{0.3} AtAirportOnTime$
- rules for causal reasoning: $FaultyPowerCord \mapsto_{0.99} DisplayOff$
- rules for diagnostic reasoning: $DisplayOff \mapsto_{0.7} SleepMode$

Issues with locality, e.g., how can 0.3 account for "all" the evidence?
Issues with combination, e.g., $FaultyPowerCord$ causes $SleepMode$?



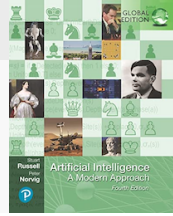
Methods for handling uncertainty II

Probability

Given the available evidence, A_{25} will get me there on time with probability 0.04

Remark. Fuzzy logic handles **degree of truth** NOT uncertainty e.g.,

- *TrafficCongested* is true to degree 0.8



Methods for handling uncertainty III

Probability versus fuzzy logic

Probability: uncertainty about a fact

$$P(Height \geq 180 \text{ cm}) = 0.7$$

We do not know Alice's height.

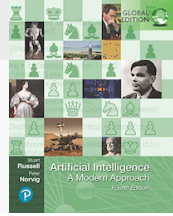
Fuzzy logic: vagueness of a concept

$$\mu_{Tall}(Alice) = 0.7$$

We know that Alice is 178 cm tall, but "tall" has no sharp boundary.

Takeaway

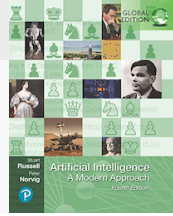
Probability models incomplete knowledge.
Fuzzy logic models concepts with gradual boundaries.



Probability I

Probabilistic assertions **summarize uncertainty arising from:**

- model simplification: we do not enumerate every exception, condition, and possible cause;
- incomplete information: relevant facts or initial conditions are not available.



Probability II

Probability can be interpreted in different ways.

Frequentist interpretation

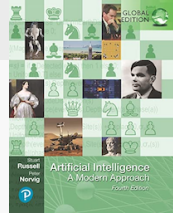
Probability is the long-run relative frequency of an event in repeated, comparable trials. Example: "In similar journeys, 6% arrived at the airport on time."

Bayesian interpretation

Probability represents an agent's degree of belief, given the evidence currently available. e.g., "Given what I know now, I assign probability 0.06 to arriving at the airport on time."

Why do we use it in this module?

An AI agent must reason about a single situation, combine evidence, and update its beliefs when new information becomes available.



Probability III

Subjective or Bayesian probability

A probability expresses an agent's degree of belief, given the evidence currently available.

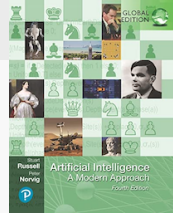
- e.g., $P(AtAirportOnTime \mid A_{25}, \text{ no reported accidents}) = 0.06$

How to interpret this value

The value 0.06 describes the agent's current degree of belief about this trip.
It is **not** an intrinsic property of the trip itself.

Past experience from similar situations can help estimate this probability.

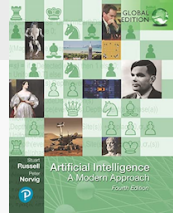
Different agents may assign different probabilities if they have different evidence or background knowledge.



Probability IV

New evidence changes the agent's belief:

- e.g.,
  $P(AtAirportOnTime \mid A_{25}, \text{ no reported accidents, 5 a.m.}) = 0.15$



Making decisions under uncertainty

Suppose I believe the following:

$$
\begin{aligned}
P(A_{25} \text{ gets me there on time} | \ldots) &= 0.04 \\
P(A_{90} \text{ gets me there on time} | \ldots) &= 0.70 \\
P(A_{120} \text{ gets me there on time} | \ldots) &= 0.95 \\
P(A_{1440} \text{ gets me there on time} | \ldots) &= 0.9999
\end{aligned}
$$

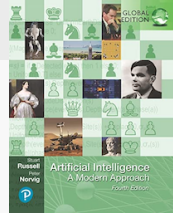
Which action to choose?

- Depends on my preferences for missing flight vs. airport cuisine, etc.
- Being Rational means following Maximum Expected Utility principle
- Utility theory is used to represent and infer preferences
- Decision theory = utility theory + probability theory

Basic probability notation

Key idea

Probability provides a formal language to describe possible worlds, events, random variables, and an agent's degree of belief in uncertain propositions.



Probability basics I

Both logic and probability reason about possible worlds:

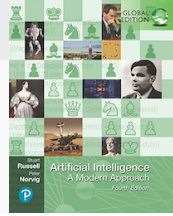
- Logical assertions distinguish worlds that are possible from worlds that are impossible;
- Probabilistic assertions assign a probability to each possible world, expressing how plausible it is.

Sample space, sample points, and events

Ω is the sample space: the set of all possible worlds.
Each $\omega \in \Omega$ is a sample point or possible world/atomic event.
An event is any subset $A \subseteq \Omega$.

A sample point $\omega \in \Omega$ is a **complete possible world**:

- Distinct possible worlds are mutually exclusive: *only one can occur*.
- The possible worlds are exhaustive: *one of them must occur*.



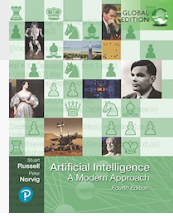
Probability basics II

Example: a roll of a fair die

$$\Omega = \{1, 2, 3, 4, 5, 6\}$$

$\omega = 3$ is one possible world (sample point)

$A = \{\omega \in \Omega \mid \omega < 4\} = \{1, 2, 3\}$ is an event



Probability basics III

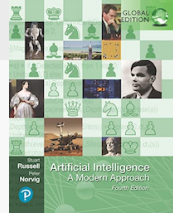
Probability space

A probability space is a triple $(\Omega, \mathcal{F}, P)$
where:

- Ω is the sample space, i.e., the set of all possible worlds;
- $\mathcal{F}$ is a σ-algebra of events, with $\mathcal{F} \subseteq 2^{\Omega}$[a];
- $P : \mathcal{F} \to [0, 1]$ is a probability measure, such that $P(\Omega) = 1$ and, for pairwise disjoint events $A_1, A_2, \ldots,$

$$P\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} P(A_i).$$

[a] $\mathcal{F}$ specifies which subsets of Ω are events to which we can assign a probability.



Probability basics IV

Probability space: discrete sample space

For a finite or countably infinite discrete sample space Ω, we take $\mathcal{F} = 2^{\Omega}$
Hence, every subset $A \subseteq \Omega$ is an event.

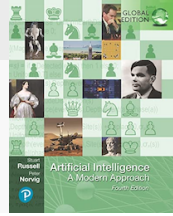
A probability model assigns a probability $P(\omega)$ to every possible world $\omega \in \Omega$, such that:

- $0 \leq P(\omega) \leq 1$
- $\sum_{\omega \in \Omega} P(\omega) = 1$

For any event $A \subseteq \Omega$:

$$P(A) = \sum_{\omega \in A} P(\omega)$$

e.g., $P(1) = P(2) = P(3) = P(4) = P(5) = P(6) = 1/6$.
$P(\text{die roll} < 4) = P(1) + P(2) + P(3) = 1/6 + 1/6 + 1/6 = 1/2$



Random variables I

Random variable

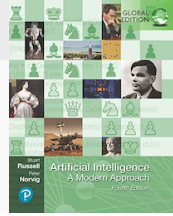
A random variable is a **function**

$$X : \Omega \to \mathcal{X},$$

which associates each possible world with a value in its range $\mathcal{X}$ (e.g., the reals or Booleans).

e.g., $Odd(1) = true$.

As you can note, the probability measure P on Ω induces a probability distribution for every random variable X.



Random variables II

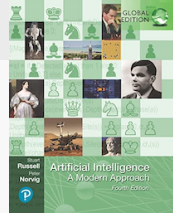
Probability distribution of a random variable

The probability distribution of X assigns a probability to every possible value $x_i \in \mathcal{X}$:

$$P_X(x_i) = P(X = x_i).$$

The distribution is normalized:

$$\sum_{x_i \in \mathcal{X}} P_X(x_i) = 1.$$



Random variables III

Example: a fair die roll

$$P(Odd = true) = P(1) + P(3) + P(5) = \frac{1}{6} + \frac{1}{6} + \frac{1}{6} = \frac{1}{2}$$

$$P(Odd = false) = P(2) + P(4) + P(6) = \frac{1}{2}$$

Therefore:

$$P_X(Odd) = \left\langle \frac{1}{2}, \frac{1}{2} \right\rangle$$

From random variables to events

A random variable can take different values across possible worlds.

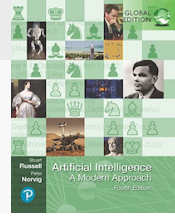
An event is specified by stating which value, or values, a random variable may take.

$$\underbrace{\mathit{Weather}}_{\text{random variable}} \longrightarrow \underbrace{\mathit{Weather} = \mathit{rain}}_{\text{event}}$$

The event $\mathit{Weather} = \mathit{rain}$ is the set of possible worlds in which the variable *Weather* takes the value *rain*.

We can therefore ask:

$$P(\mathit{Weather} = \mathit{rain}).$$



Propositions as events I

Propositions as events

A proposition denotes the event consisting of all possible worlds in which it is true.

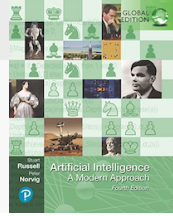
Let A and B be Boolean random variables. We use:

$$a \equiv (A = true) \qquad b \equiv (B = true)$$

Each sample point is a complete truth assignment:

$$\Omega = \{a \wedge b,\ a \wedge \neg b,\ \neg a \wedge b,\ \neg a \wedge \neg b\}.$$

Proposition	Atomic events where it is true
a	$a \wedge b,\ a \wedge \neg b$
$\neg a$	$\neg a \wedge b,\ \neg a \wedge \neg b$
$a \wedge b$	$a \wedge b$
$a \vee b$	$a \wedge b,\ a \wedge \neg b,\ \neg a \wedge b$

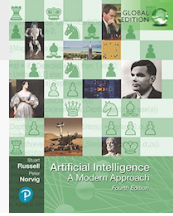


Propositions as events II

Therefore, a proposition is a disjunction of mutually exclusive atomic events:

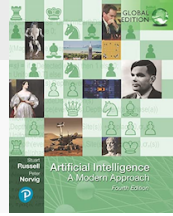
$$a \vee b \equiv (a \wedge b) \vee (a \wedge \neg b) \vee (\neg a \wedge b)$$

$$P(a \vee b) = P(a \wedge b) + P(a \wedge \neg b) + P(\neg a \wedge b).$$



Syntax for propositions

- Propositional or Boolean random variables
  e.g., *Cavity* (do I have a cavity?)
  $Cavity = true$ is a proposition, also written *cavity*
- Discrete random variables (finite or infinite)
  e.g., *Weather* is one of $\langle sunny, rain, cloudy, snow \rangle$
  $Weather = rain$ is a proposition
  Values must be exhaustive and mutually exclusive
- Continuous random variables (bounded or unbounded)
  e.g., $Temp = 21.6$; $Temp < 22.0$
- Arbitrary Boolean combinations of basic propositions

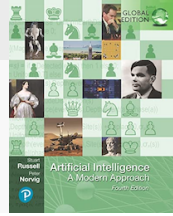


Examples of propositions

Each basic proposition identifies the event in which a random variable takes a specified value, or satisfies a specified condition.

Variable type	Proposition	Interpretation
Boolean	$Cavity = true$, also written $cavity$	"The patient has a cavity"
Discrete	$Weather = rain$	"The weather is rainy"
Continuous	$Temp < 22.0$	"The temperature is below 22°C"

Each proposition corresponds to an event and can therefore be assigned a probability.



Probability distributions and propositions I

Discrete random variable

Let $\mathcal{W} = \{sunny, cloudy, rainy, snowy\}$ be the domain of the discrete random variable *Weather*.
Its probability distribution assigns a probability to each possible value:

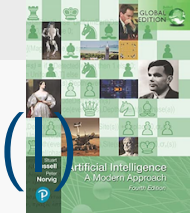
$$P_{Weather} = \langle 0.72,\ 0.10,\ 0.08,\ 0.10 \rangle.$$

That is:

$$P(Weather = sunny) = 0.72, \quad P(Weather = cloudy) = 0.10,$$
$$P(Weather = rainy) = 0.08, \quad P(Weather = snowy) = 0.10.$$

The distribution is normalized:

$$\sum_{w \in \mathcal{W}} P(Weather = w) = 1.$$



Continuous r.v. first intuition: a point versus a region (I)

Dartboard

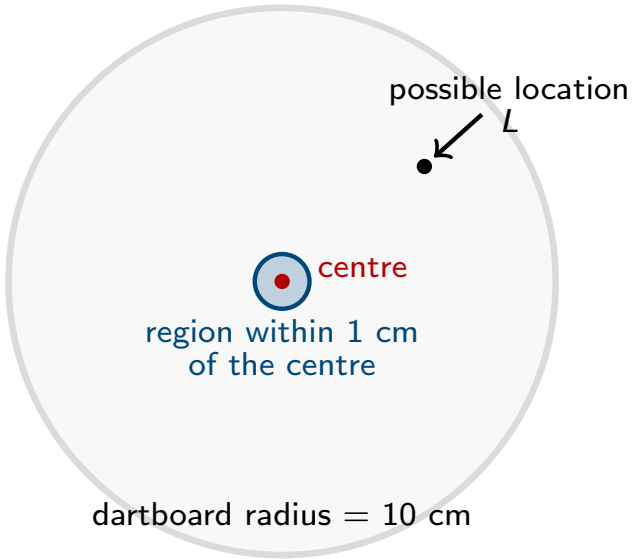
A dart lands uniformly at random on a circular dartboard of radius 10 cm.

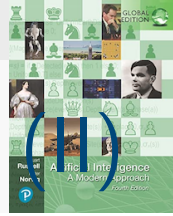
Let L denote the location where the dart lands. The dartboard is a continuous space: there are infinitely many possible landing locations.

The centre is one exact location, i.e., a single point with no area:

$$P(L = \text{centre}) = 0.$$

This does **not** mean that hitting the centre is impossible. It means that an exact point is too small to have positive probability in a continuous space.





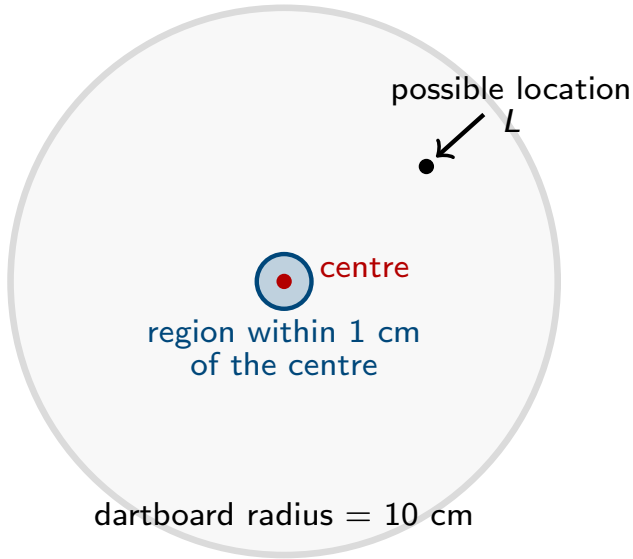
Continuous r.v. a first intuition: a point versus a region (I)

Dartboard

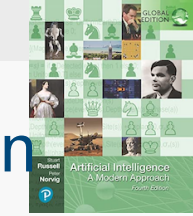
A dart lands uniformly at random on a circular dartboard of radius 10 cm.

To ask a meaningful probability question, we consider a region around the point. For example, the region within 1 cm of the centre has positive area:

$$P(\text{distance}(L, \text{centre}) \leq 1) = \frac{\pi \cdot 1^2}{\pi \cdot 10^2} = 0.01.$$



For a continuous random variable, probabilities concern regions of possible values, rather than exact individual values.



Continuous r.v. a first intuition: a point versus a region (III)

How late does a student arrive?

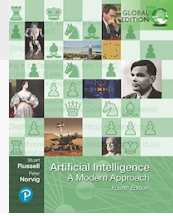
Let T denote the number of minutes by which a student arrives late.

With a clock that records whole minutes, we may observe that the student arrives during the fifth minute:

$$5 \leq T \leq 6.$$

As the measurement becomes more precise, the interval around 5 minutes becomes narrower. In the ideal continuous model:

$$P(T = 5) = 0, \qquad P(5 \leq T \leq 6) > 0.$$



Probability for continuous variables I

From point probabilities to probability densities

A probability density function describes how probability is locally distributed over the possible values of the variable.

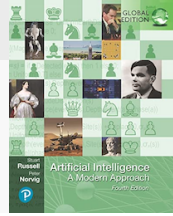
Probability density function

Let X be a continuous random variable with values in $\mathcal{X}$. A nonnegative, integrable function

$$p_X : \mathcal{X} \to \mathbb{R}_{\geq 0}$$

is a probability density function (pdf) for X if:

$$\int_{\mathcal{X}} p_X(x)\, dx = 1.$$



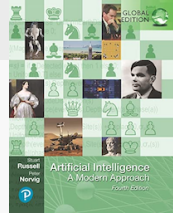
Probability for continuous variables II

The probability that X falls within an interval is the area under its density curve: $P(a \leq X \leq b) = \int_a^b p_X(x)\, dx$.

Important distinction

$p_X(x)$ is a **density**, not the probability of the exact value x. For a continuous variable:

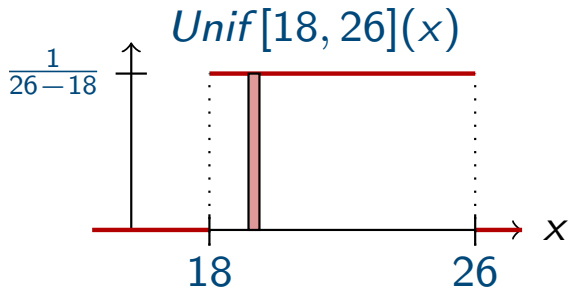
$$P(X = x) = 0.$$



Probability for continuous variables: examples I

A common pdf is the uniform distribution. If $X \sim \mathit{Unif}[a, b]$, then:

$$p_X(x) = \begin{cases} \dfrac{1}{b-a} & \text{if } a \leq x \leq b, \\ 0 & \text{otherwise.} \end{cases}$$

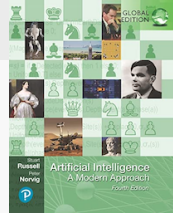


For example, if $X \sim \mathit{Unif}[18, 26]$:

$$p_X(20.5) = \frac{1}{26-18} = 0.125$$

The density value 0.125 describes the probability in a small interval around 20.5:

$$P(20.5 \leq X \leq 20.5 + \Delta x) = 0.125\,\Delta x.$$



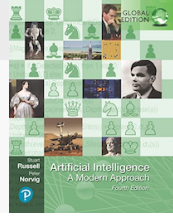
Probability for continuous variables: examples II

A common notational convention

Russell and Norvig, like some other AI texts, use $P(X = x)$, even when X is a continuous random variable, to denote a density value. Strictly speaking, this is an **abuse of notation**.

Thus, $P(X = 20.5) = 0.125$ is intended to mean:

$$\lim_{\Delta x \to 0} \frac{P(20.5 \leq X \leq 20.5 + \Delta x)}{\Delta x} = 0.125.$$



Probability for continuous variables: examples III

Another common pdf is the Gaussian (Normal) distribution . If

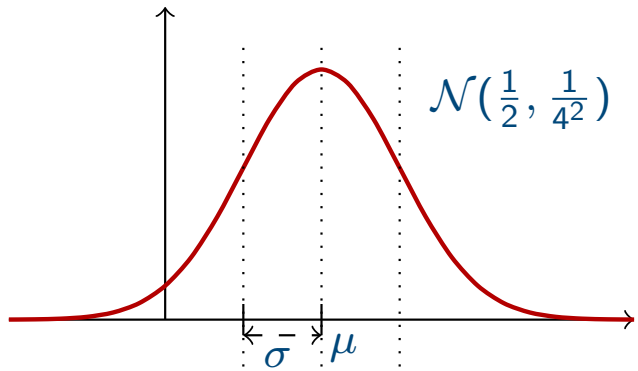
$$X \sim \mathcal{N}(\mu, \sigma^2),$$

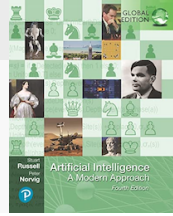
then its probability density function is:

$$p_X(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{\frac{-(x-\mu)^2}{2\sigma^2}}$$

where, μ is the mean and σ is the standard deviation. The standard Gaussian has mean 0 and variance 1:

$$p_X(x) = \frac{1}{\sqrt{2\pi}} e^{\frac{-x^2}{2}}$$





Prior probability

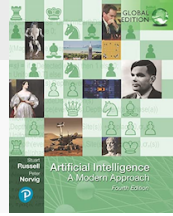
Prior probability

The prior or unconditional probability of a proposition expresses an agent's belief before incorporating new evidence.

Example. Before observing the current weather conditions, the agent has the following prior distribution:

$$P_{Weather} = \langle 0.72,\ 0.10,\ 0.08,\ 0.10 \rangle,$$

New evidence allows the agent to update this prior distribution and obtain a posterior distribution.



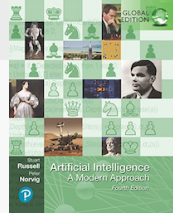
Joint probability distribution I

Joint probability distribution

The joint probability distribution over a set of random variables assigns a probability to every complete assignment of values to those variables.

Example. The joint distribution $P(Weather, Cavity)$ is a 4×2 table. Each entry is the probability of one atomic event, such as $Weather = sunny \wedge Cavity = true$:

$Weather =$	*sunny*	*rainy*	*cloudy*	*snowy*
$Cavity = true$	0.144	0.020	0.016	0.020
$Cavity = false$	0.576	0.080	0.064	0.080

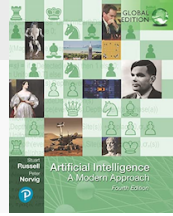


Joint probability distribution II

Every event involving *Weather* and *Cavity* is a set of atomic events. Its probability is obtained by summing the corresponding entries of the joint distribution.

Example. Answering a query from the joint distribution The event *Weather* = *sunny* includes two atomic events:

$$\begin{aligned} P(\mathit{Weather} = \mathit{sunny}) &= P(\mathit{Weather} = \mathit{sunny} \wedge \mathit{Cavity} = \mathit{true}) \\ &\quad + P(\mathit{Weather} = \mathit{sunny} \wedge \mathit{Cavity} = \mathit{false}) \\ &= 0.144 + 0.576 = 0.72. \end{aligned}$$

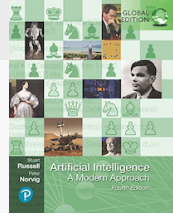


Conditional probability I

Conditional and posterior probability

A conditional probability $P(X \mid Evidence = e)$ represents an agent's belief about X, given that $Evidence = e$ is known.

When $Evidence = e$ is newly observed evidence, $P(X \mid Evidence = e)$ is called a posterior probability. The corresponding $P(X)$, before observing the evidence, is the prior probability.



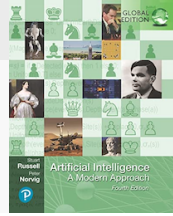
Conditional probability II

Example.

$$P(Cavity = true \mid Toothache = true) = 0.8.$$

This means: given that the patient has a toothache, the agent assigns probability 0.8 to the proposition that the patient has a cavity.
It is **not a logical or causal rule** of the form:

$$Toothache \implies Cavity.$$



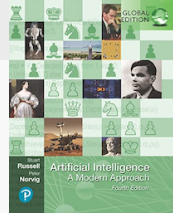
Conditional probability III

Notation for sets of conditional distributions: $\mathbf{P}(Cavity \mid Toothache)$ is a 2-element vector of 2-element vectors.

Conditional distributions

For every possible value of *Toothache*, the expression $P(Cavity \mid Toothache)$ specifies a probability distribution over the possible values of *Cavity*. This conditional distribution can be represented as a 2×2 table:

	$Cavity = true$	$Cavity = false$
$Toothache = true$	0.8	0.2
$Toothache = false$	0.1	0.9



Conditional probability IV

Suppose that the only available evidence is *toothache*:

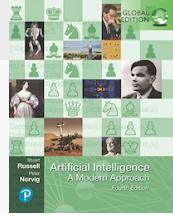
$$P(cavity \mid toothache) = 0.8.$$

If the agent subsequently observes that the patient has a cavity, its updated belief is:

$$P(cavity \mid toothache, cavity) = 1.$$

More evidence

The original value $P(cavity \mid toothache) = 0.8$ remains a correct statement about the agent's belief when only *toothache* is known. Once *cavity* is also observed, however, it is **no longer the most useful** probability for the agent's current state of knowledge.

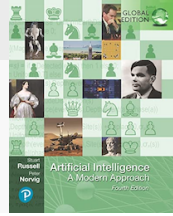


Conditional probability V

Some additional evidence may be irrelevant. For example, if domain knowledge tells us that 49*ersWin* is irrelevant to cavities once *toothache* is known, then:

$$P(cavity \mid toothache, 49ersWin) = P(cavity \mid toothache) = 0.8.$$

This form of simplification is based on conditional independence, which we will discuss later.



Conditional probability VI

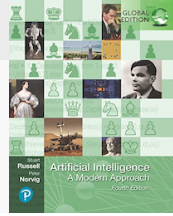
Conditional probability

For two propositions a and b, with $P(b) \neq 0$:

$$P(a \mid b) = \frac{P(a \wedge b)}{P(b)}.$$

The product rule follows immediately:

$$P(a \wedge b) = P(a \mid b)\, P(b) = P(b \mid a)\, P(a).$$



From conditional to joint probability I

The product rule

For two propositions a and b:

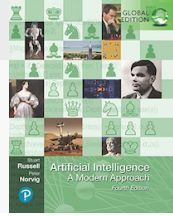
$$P(a \wedge b) = P(a \mid b)\, P(b).$$

The rule tells us how to obtain the probability that two propositions are both true:

1. start from the probability of b;
2. consider the probability of a, given that b is true;
3. multiply the two values.

Example.

$$P(Cavity = true \wedge Toothache = true)$$
$$= P(Cavity = true \mid Toothache = true)\, P(Toothache = true).$$



From conditional to joint probability II

By applying the product rule to every combination of values of two random variables, we can construct their joint probability distribution.

Example. For $Weather = sunny$ and $Cavity = true$:

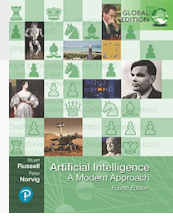
$$P(Weather = sunny, Cavity = true) = \\ = P(Weather = sunny \mid Cavity = true) \cdot P(Cavity = true).$$

Repeating this calculation for every possible pair

$$(w, c) \in \{sunny, rainy, cloudy, snowy\} \times \{true, false\}$$

produces all $4 \times 2 = 8$ entries of $P(Weather, Cavity)$.

A joint distribution represents the probabilities of all complete assignments to the variables in the domain.



From the product rule to the chain rule I

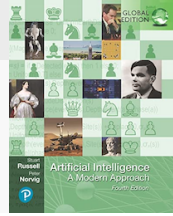
Three random variables

For three random variables, apply the product rule first to $P(X_1, X_2, X_3)$:

$$P(X_1, X_2, X_3) = P(X_1, X_2)\, P(X_3 \mid X_1, X_2).$$

Now apply the product rule again to $P(X_1, X_2)$:

$$P(X_1, X_2, X_3) = P(X_1)\, P(X_2 \mid X_1) \cdot P(X_3 \mid X_1, X_2).$$



From the product rule to the chain rule II

The same reasoning can be repeated for any number of variables. The Chain rule is derived by successive application of product rule:

$$\begin{aligned}\mathbf{P}(X_1,\ldots,X_n) &= \mathbf{P}(X_1,\ldots,X_{n-1})\,\mathbf{P}(X_n|X_1,\ldots,X_{n-1}) \\ &= \mathbf{P}(X_1,\ldots,X_{n-2})\,\mathbf{P}(X_{n-1}|X_1,\ldots,X_{n-2})\,\mathbf{P}(X_n|X_1,\ldots,X_{n-1}) \\ &= \ldots \\ &= \textstyle\prod_{i=1}^{n}\mathbf{P}(X_i|X_1,\ldots,X_{i-1})\end{aligned}$$

Chain rule

$$P(X_1,\ldots,X_n) = \prod_{i=1}^{n} P(X_i \mid X_1,\ldots,X_{i-1}).$$

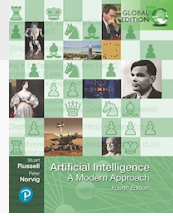
For $i = 1$, the conditioning set is empty:

$$P(X_1 \mid \{\}) = P(X_1).$$

Inference using full joint distributions

Key idea

If we know a complete joint probability distribution, we can answer any probability query by summing the entries corresponding to the possible worlds in which the query is true.



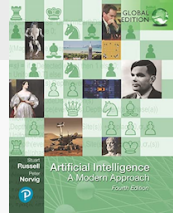
Inference by enumeration

Start with the joint distribution:

	toothache		$\neg$*toothache*	
	catch	$\neg$*catch*	*catch*	$\neg$*catch*
cavity	0.108	0.012	0.072	0.008
$\neg$*cavity*	0.016	0.064	0.144	0.576

For any proposition ϕ, sum the atomic events where it is true:
$P(\phi) = \sum_{\omega:\omega \models \phi} P(\omega)$

e.g. $\phi = \mathit{toothache}$



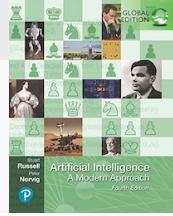
Inference by enumeration

Start with the joint distribution:

	toothache		$\neg$*toothache*	
	catch	$\neg$*catch*	*catch*	$\neg$*catch*
cavity	0.108	0.012	0.072	0.008
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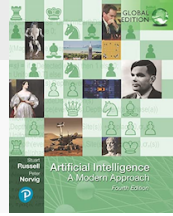
Inference by enumeration

Start with the joint distribution:

	toothache		¬*toothache*	
	catch	¬*catch*	*catch*	¬*catch*
cavity	0.108	0.012	0.072	0.008
¬*cavity*	0.016	0.064	0.144	0.576

For any proposition ϕ, sum the atomic events where it is true:
$P(\phi) = \sum_{\omega:\omega \models \phi} P(\omega)$

$P(toothache) = 0.108 + 0.012 + 0.016 + 0.064 = 0.2$



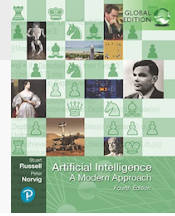
Inference by enumeration

Start with the joint distribution:

	toothache		¬*toothache*	
	catch	¬*catch*	*catch*	¬*catch*
cavity	0.108	0.012	0.072	0.008
¬*cavity*	0.016	0.064	0.144	0.576

For any proposition ϕ, sum the atomic events where it is true:
$P(\phi) = \sum_{\omega:\omega \models \phi} P(\omega)$

$P(toothache) = 0.108 + 0.012 + 0.016 + 0.064 = 0.2$
Answer a new query: e.g., $\phi = cavity \vee toothache$



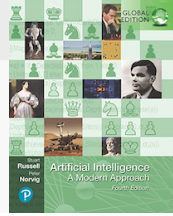
Inference by enumeration

Start with the joint distribution:

	toothache		¬*toothache*	
	catch	¬*catch*	*catch*	¬*catch*
cavity	0.108	0.012	0.072	0.008
¬*cavity*	0.016	0.064	0.144	0.576

For any proposition ϕ, sum the atomic events where it is true:
$P(\phi) = \sum_{\omega:\omega \models \phi} P(\omega)$

$P(toothache) = 0.108 + 0.012 + 0.016 + 0.064 = 0.2$
$P(cavity \vee toothache) = 0.108+0.012+0.072+0.008+0.016+0.064 = 0.28$



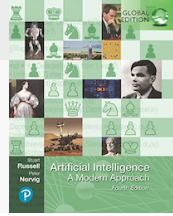
Inference by enumeration

Start with the joint distribution:

	toothache		$\neg$*toothache*	
	catch	$\neg$*catch*	*catch*	$\neg$*catch*
cavity	0.108	0.012	0.072	0.008
$\neg$*cavity*	0.016	0.064	0.144	0.576

Can also compute conditional probabilities:

$$P(\neg cavity \mid toothache) = \frac{P(\neg cavity \wedge toothache)}{P(toothache)}$$



Inference by enumeration

Start with the joint distribution:

	toothache		¬toothache	
	catch	¬*catch*	*catch*	¬*catch*
cavity	0.108	0.012	0.072	0.008
¬*cavity*	0.016	0.064	0.144	0.576

Can also compute conditional probabilities:

$$P(\neg cavity \mid toothache) = \frac{P(\neg cavity \wedge toothache)}{P(toothache)}$$
$$= \frac{0.016 + 0.064}{0.108 + 0.012 + 0.016 + 0.064} = 0.4$$

Conditioning restricts the possible worlds

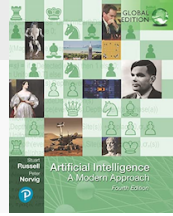
The complete joint distribution assigns total probability 1 to all possible worlds. Suppose that we observe *toothache* = *true*. We retain only the possible worlds compatible with this evidence:

$$P(toothache) = 0.108 + 0.012 + 0.016 + 0.064 = 0.20.$$

What does 0.20 mean?

Before conditioning, the worlds in which the patient has a toothache carry a total probability mass of 0.20.

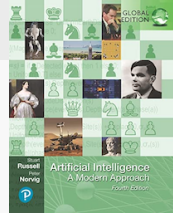
The remaining mass, 0.80, belongs to worlds that are incompatible with the observed evidence and is excluded from the conditional analysis.



Normalization I

Within the worlds compatible with $toothache = true$:

$$\begin{aligned} P(Cavity, toothache) &= \langle P(cavity, toothache), P(\neg cavity, toothache)\rangle \\ &= \langle 0.108 + 0.012,\ 0.016 + 0.064\rangle \\ &= \langle 0.12,\ 0.08\rangle. \end{aligned}$$



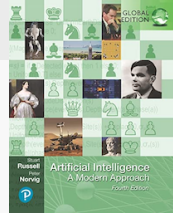
Normalization II

Why normalize?

The vector $\langle 0.12, 0.08 \rangle$ contains the probability mass of the two alternatives in the original joint distribution. Its sum is $P(toothache) = 0.20$.

Once *toothache* is observed, we redistribute this mass over the two remaining alternatives so that their probabilities sum to 1, while preserving their relative weights.

$$\begin{aligned} P(Cavity \mid toothache) &= \frac{1}{P(toothache)} P(Cavity, toothache) \\ &= \frac{1}{0.20} \langle 0.12,\ 0.08 \rangle \\ &= \langle 0.6,\ 0.4 \rangle. \end{aligned}$$



Normalization: a compact form

The normalization factor can be denoted by α:

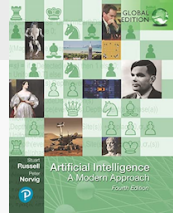
$$\alpha = \frac{1}{P(toothache)} = \frac{1}{0.20} = 5.$$

Using α, the calculation from the previous slide becomes:

$$\begin{aligned} P(Cavity \mid toothache) &= \alpha\, P(Cavity, toothache) \\ &= \alpha\, \langle P(cavity, toothache),\ P(\neg cavity, toothache) \rangle \\ &= \alpha\, \langle 0.108 + 0.012,\ 0.016 + 0.064 \rangle \\ &= 5 \langle 0.12,\ 0.08 \rangle \\ &= \langle 0.6,\ 0.4 \rangle. \end{aligned}$$

Why do we sum over *Catch*?

Catch is neither observed nor queried. We therefore sum over its possible values, while keeping *Cavity* in the result.

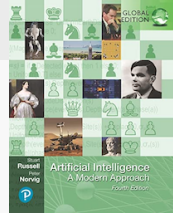


Conditional inference by enumeration

Conditional inference is enumeration of possible worlds with evidence

1. Fix the observed evidence $E = e$.
2. For each possible value of Y, sum the probabilities of all compatible worlds, varying the hidden variables H.
3. Normalize the resulting values using α, so that the distribution over Y sums to 1.

As for any probability query, we sum the probabilities of compatible possible worlds. Conditioning additionally fixes the evidence and normalizes the final distribution.



Operations on a joint probability distribution I

Marginalization, or summing out

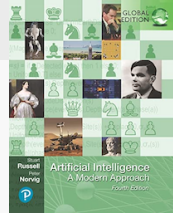
To obtain the distribution of one variable, sum over the possible values of the other variable.

Start with the joint distribution:

	Weather			
Cavity	*sunny*	*rain*	*cloudy*	*snow*
true	0.144	0.020	0.016	0.020
false	0.576	0.080	0.064	0.080

For example, to compute $P(Weather)$, we sum out $Cavity$:

$$\begin{aligned} P(Weather = sunny) &= P(sunny, cavity) + P(sunny, \neg cavity) \\ &= 0.144 + 0.576 = 0.72. \end{aligned}$$



Operations on a joint probability distribution II

Conditioning

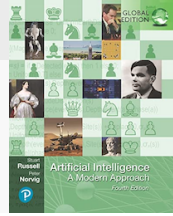
To compute a distribution given observed evidence, retain only the entries compatible with that evidence and normalize them.

Start again from the same joint distribution:

	Weather			
Cavity	*sunny*	*rain*	*cloudy*	*snow*
true	0.144	0.020	0.016	0.020
false	0.576	0.080	0.064	0.080

For example, after observing $Weather = sunny$:

$$\begin{aligned} P(Cavity \mid Weather = sunny) &= \frac{1}{P(Weather = sunny)} \langle 0.144,\ 0.576 \rangle \\ &= \frac{1}{0.72} \langle 0.144,\ 0.576 \rangle \\ &= \langle 0.2,\ 0.8 \rangle. \end{aligned}$$



Probability queries I

Probability query

A probability query asks for the agent's posterior joint probability distribution over one or more query variables Y, given observed values $E = e$ of some evidence variables:

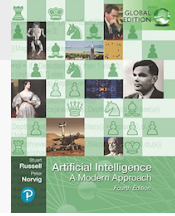
$$P(Y \mid E = e).$$

Let X denote all random variables in the domain. We distinguish three disjoint sets of variables:

- Y: query variables, whose posterior distribution we seek;
- E: evidence variables, whose values e have been observed;
- $H = X \setminus (Y \cup E)$: hidden variables.

Example

In $P(Cavity \mid Toothache = true)$: $Y = Cavity$, $E = Toothache$, $H = Catch$.



Probability queries II

Inference by enumeration

To answer $P(Y \mid E = e)$, fix the evidence and sum over every possible assignment h of the hidden variables:

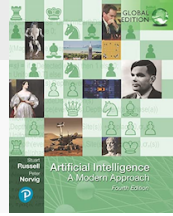
$$P(Y \mid E = e) = \alpha\, P(Y, E = e) = \alpha \sum_h P(Y, E = e, H = h).$$

The terms $P(Y, E = e, H = h)$ are entries of the full joint distribution, because Y, E, and H together include all variables in the domain.

The constant $\alpha = \frac{1}{P(E=e)}$ normalizes the resulting distribution over Y, so it sums to 1.

Connection with the previous example

For $P(\mathit{Cavity} \mid \mathit{Toothache} = \mathit{true})$, we fix $\mathit{Toothache} = \mathit{true}$, sum over the possible values of *Catch*, and normalize the resulting values for *Cavity*.



Probability queries III

Why full enumeration does not scale

Suppose that the domain contains n random variables and that each variable has at most d possible values.

- The full joint distribution may contain up to d^n entries.
- Storing it requires $O(d^n)$ space.
- Enumerating and summing its entries requires $O(d^n)$ time in the worst case.
- We must also determine the probability associated with every entry.

Next question

How can we represent a joint probability distribution compactly and perform inference without explicitly listing all possible worlds?

Questions?

